

# Dewan Mahnaaz Mahmud

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## Employment

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- **Lecturer, Department of Automatic Control and Robotics** Nov 2024-Ongoing  
*École Centrale de Nantes, France*
  - Courses taught: Smart grids for renewable energy (Winter 2026), Signals, systems, linear drive (Fall 2025), Control of electrical drive systems (Fall 2024, Fall 2025), Power systems dynamics (Fall 2024, Fall 2025), Power grid control (Fall 2024)
- **Internship, Laboratoire des Sciences du numerique de Nantes (LS2N)** Nov 2024-Ongoing  
*École Centrale de Nantes, France*
  - Master thesis: Renewables integration to secondary voltage control

## Education

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- **Ph.D., Automatic Control** Nov 2024-Ongoing  
*École Centrale de Nantes, France*
  - Dissertation: *Higher Penetration of Renewable Energy Sources using Grid-forming Inverters*
  - Advisors: Mickaël Hilairet, Bogdan Marinescu, Vinu Thomas
- **European Master in Dynamics of Renewables-based Power Systems** Sep 2022-Aug 2024  
*École Centrale de Nantes, Universitat Politècnica de Catalunya*
  - Degrees issued: M.Sc., Automatic, robotics, M.Sc., Energy Engineering
  - GPA: 3.49/4.0
  - Master thesis: Renewables integration to secondary voltage control
  - Specialization: Power electronics in power systems
- **B.Sc., Electrical and Electronics Engineering** Apr 2017-Oct 2022  
*American International University-Bangladesh*
  - GPA: 3.78/4.00
  - Capstone project: Design and simulation of modern electric vehicle with wireless charging capability
  - Specialization: Renewable energy technology, Power system protection

## Research Experience

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- **École Centrale de Nantes** Sep 2022–Present
  - **Ph.D. Thesis: Higher penetration of renewable energy sources using grid-forming inverters**
    - Reviewed state-of-the-art methods to improve the performance of grid-forming inverters.
    - Investigating the performance of grid-forming inverters under transient conditions.
    - EMT modelling and real-time simulation of grid-forming inverters.
    - Validation of control schemes using a Hardware-in-the-Loop (HiL) experimental testbench.
  - **M.Sc. thesis: Renewables integration to secondary voltage control**
    - Investigating the participation of inverter-based resources in the secondary voltage control framework.
    - Proposed a model-free control strategy using an intelligent proportional–integral (iPI) controller for grid-connected inverters.
    - Enabled renewable energy sources to participate in secondary voltage control.
    - Validation of the iPI controller using a Hardware-in-the-Loop (HiL) experimental testbench.
  - **Semester project: Active and reactive power control of a grid-connected solar photovoltaic farm**
    - Designed a grid-following control strategy for active and reactive power control of 1 MW grid-connected solar PV.
    - Validated the components in MATLAB/Simulink® and studied the dynamics.
    - Performed case studies to validate the control strategy.
- **Universitat Politècnica de Catalunya** Feb 2023–July 2023
  - **Semester project: Design of grid-forming converter for modern power systems with high penetration of power electronics**

- Modelling of grid-forming converter for low inertia power systems.
- Validated control strategies with two different power synchronization loops.
- Designed a SISO transfer function model of the system with validation.

## Technical Skills

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MATLAB, MATLAB/Simulink®, PSCAD, DigSILENT PowerFactory, RTLab, NEPLAN, PSAT, HOMER, PSIM, Python, C++, NI Circuit Design Suite, LaTeX, Microsoft Office Suite

## Publications

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- Mahmud, D.M.; Thomas, V.; Marinescu, B.; “**Model-Free Control for Multi-Time Scale Dynamics of Grid-Connected Power Converters**” DOI: <https://doi.org/10.48550/arXiv.2606.17873>
- Mahmud, D.M.; Thomas, V.; Marinescu, B.; Hilaiet, M.; “**Saturation Feedback-Based Droop Control for Synchronization Stability Enhancement of Grid-Forming Inverters Under Frequency Dips**” *2026 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*, accepted.
- Mahmud, D.M.; Thomas, V.; Marinescu, B.; Hilaiet, M.; “**Critical Clearing Time Enhancement of Droop-Controlled Grid-Forming Inverters with Adaptive Function-Based Parameters**” *2026 IEEE Energy Conversion Congress and Exposition Europe (ECCE Europe)*, accepted. DOI: <https://doi.org/10.48550/arXiv.2603.01831>
- Mahmud, D.M.; Sule, E.E.; Weng, T.; Thomas, V.; “**Towards clean energy: design and integration strategy of photovoltaics for residential electrification**” *Journal of Engineering Research*. DOI: <https://doi.org/10.1016/j.jer.2024.03.008>
- Mahmud, D.M.; Ahmed, S.M.M.; Hasan, S.; Zeyad, M.; “**Grid-connected microgrid: design and feasibility analysis for a local community in Bangladesh**” *Clean Energy*, Volume 6, Issue 3, June 2022, Pages 447–459, Oxford University Press. DOI: <https://doi.org/10.1093/ce/zkac022>
- List of all publications in Google Scholar: <https://tinyurl.com/2p8e637r>

## Distinctions

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- Erasmus+ Erasmus Mundus Joint Master (EMJM) Scholarship for *Master’s in Dynamics of Renewables-based Power Systems (DREAM)*
- “**High-Impact Research Category/Collection for H1 2022**”, “**Featured Article**”, “**Cover Page of the issue**” and “**Editor’s Choice**” of the journal *Clean Energy*, for the article “*Grid-connected microgrid: design and feasibility analysis for a local community in Bangladesh*” Volume 6 Issue 3, June 2022, Oxford University Press
- **Dean’s List Honor** for “*Academic Excellence*”, issued by Faculty of Electrical and Electronics Engineering (EEE), American International University-Bangladesh (AIUB)

## Leadership Experience

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- **Treasurer**—IEEE AIUB Student Branch, American International University-Bangladesh (Feb 2020–Jun 2021)
- **Section Ambassador, IEEE Day 2019**—IEEE AIUB Student Branch, American International University-Bangladesh (Jul 2019–Feb 2020)
- **Event Designer**—IEEE AIUB Student Branch, American International University-Bangladesh (Jan 2019–Feb 2020)

## Presentations and Invited Talks

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- Presented research outcomes at the *IEEE Region 10 Humanitarian Technology Conference (IEEE R10-HTC 2021)*.
- Presented at the Winter School of the *European Master in Dynamics of Renewables-based Power Systems (DREAM)*, organized by *École Centrale de Nantes*.
- Conducted a webinar as a speaker titled “*AIUBians on Erasmus*”, organized by the *IEEE AIUB Student Branch*.

## Language

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English (Fluent (C1)), French (Beginner (A2)), Bengali (Native)